

Study of moist air transfer and its thermal load under refrigerator door closing and opening conditions

Guixiang He^a, Huawei Zhang^{a,b}, Guoqiang Liu^{a,*}, Tianyang Zhao^a, Gang Yan^a

^a Department of Refrigeration and Cryogenic Engineering, School of Energy and Power Engineering, Xi'an Jiaotong University, 710049, Xi'an, China

^b Hefei Hualing Co., Ltd, 230601, Hefei, China

ARTICLE INFO

Keywords:

Refrigerator
Moist air transfer
Door opening-closing
Thermal load

ABSTRACT

This study conducted a detailed experimental study of moist air transfer and its thermal loads in refrigerators under door closing and opening conditions. Employing trace gas technology with CO₂ as the tracer, a mass balance equation for CO₂ was formulated to determine the air transfer rate. It was founded that under door closing conditions, the primary pathways for moist air transfer were the gasket and defrosting drainage tube. The proportion occupied by the defrosting drainage tube was closely related to its structural design, reaching from 29.5 % to 88.5 %. The mass transfer rates in the pull-down and intermittent refrigeration processes were 133.018 g h⁻¹ and 102.920 g h⁻¹. Under door opening conditions, the averaged mass transfer rate during the opening-closing process is observed to range from 65.598 to 150.076 g s⁻¹, approximately 2300–5200 times higher than the door closing conditions. Over a 24 h operation period, the accumulative thermal load caused by moist air transfer was 417.054 kJ, with the pull-down, intermittent refrigeration, and door opening-closing processes contributing 5.7 %, 40.2 %, and 54.1 %. These findings provide essential data support for understanding the thermodynamic behavior of moist air transfer within refrigeration systems under varying door operation conditions.

Nomenclature

v	volumetric charging rate, m ³ ·h ⁻¹	Q_{sen}	sensible thermal load, W
V	volume, m ³	T_{in}	refrigerator internal temperature, °C
c	real time tracer gas concentration, ppm	T_{out}	environment air temperature, °C
c_{atm}	background tracer gas concentration, ppm	T_{eva}	evaporator surface temperature, °C
c_0	initial tracer gas concentration, ppm	d_{in}	refrigerator internal moisture content, g·(kg _{dry air}) ⁻¹
$\dot{V}$	volume transfer rate, m ³ ·h ⁻¹	d_{out}	environment moisture content, g·(kg _{dry air}) ⁻¹
I	transfer rate, h ⁻¹	r_{wat}	latent heat of vaporization of water, kJ·kg ⁻¹
$\dot{m}$	mass transfer rate, g·h ⁻¹	r_{ice}	latent heat of fusion of ice, kJ·kg ⁻¹
ρ	air density, kg·m ⁻³	Q_{wall}	thermal load of the cabinet walls, kJ
ρ_{dry}	dry air density, kg·m ⁻³	Q_{gasket}	thermal load of the gasket, kJ
m_d	mass of water vapor, g	A	area of the wall, m ²
$\bar{c}_{p,a}$	average specific heat capacity, kJ·kg ⁻¹ ·K ⁻¹	$h_{wall,out}$	outer wall convective heat transfer coefficient, W·m ⁻² ·K ⁻¹

(continued on next page)

* Corresponding author.

E-mail address: guoqiangliu@mail.xjtu.edu.cn (G. Liu).

<https://doi.org/10.1016/j.csite.2025.105909>

Received 15 July 2024; Received in revised form 21 October 2024; Accepted 15 February 2025

Available online 18 February 2025

2214-157X/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

(continued)

Nomenclature			
t	time, h	$h_{\text{wall, in}}$	inner wall convective heat transfer coefficient, $\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$
Q_{lat}	latent thermal load, W	λ_i	thermal conductivity of different materials, $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$
Q	thermal load, W	δ_i	thickness of the wall, mm
E_{sen}	sensible cold consumption, kJ	h_{gasket}	heat transfer coefficient per unit length of the gasket, $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$
E_{lat}	latent cold consumption, kJ	L	length of the gasket, m
E	cold consumption, kJ		

1. Introduction

With the advancement of modern living standards, the demand for refrigeration appliances such as domestic refrigerators, commercial display cabinets, and large freezers has surged. Globally, it is estimated that approximately 2.3 billion refrigerators are in use, contributing to roughly 4 % of the overall electricity consumption. (UNEP, 2019) [1]. Amidst this growth, the refrigeration industry is focused on enhancing energy efficiency through measures including the adoption of environmentally friendly refrigerants [2,3], optimization of system components [4–7] and development of other advanced refrigeration technologies [8,9]. In addition to improving refrigeration systems and components during production phase for developers, it is equally important to explore the potential for energy conservation during the usage phase of refrigerators for customers. The most common action for customers during usage phase is to open and close the door. During the door opening-closing process, the internal and external air exchanges violently, causing a strong disturbance to the air temperature and humidity inside the refrigerator, which will have a negative impact on the energy consumption and food safety of the refrigerator [10,11]. Thus, it necessary to obtain the moist air transfers characteristics and its thermal load under refrigerators door closing and opening conditions.

For refrigerators, the internal air is maintained at a lower temperature. The transfer of air between interior of the cabinet and the external environment will generate sensible thermal loads. In addition, after the moist air enters the compartment, it can cause condensation or frosting on liners of refrigerated cabinets and evaporator, thereby introducing latent thermal loads [12,13]. Consequently, energy conservation can be achieved by minimizing the mass transfer of moist air. Moist air transfer is driven by pressure differentials between the air inside and outside of the cabinet. Direct measurement of air flow rates at low velocities is challenging and may disrupt the flow field. Techniques such as hot-wire anemometers [14] and pressure sensors [15], which are adept at measuring air velocity and pressure, often assumed that the flow field is uniformly distributed. However, this assumption overlooks edge effects, thereby introducing measurement errors. Non-invasive quantitative techniques such as tracer gas methods, laser doppler anemometry, and particle image velocimetry, which have little interference with the flow field, have been used to study airflow behavior during forced air cooling, refrigeration, and drying [16,17]. Among various techniques, tracer gas method is often used to measure wet air infiltration/leakage flow rates because of its simple system and high accuracy [18], and is applicable for measuring the moisture air transfer rate in refrigeration equipment. Mazyar Amin et al. [19] utilized this technique in their study of open refrigerated display cases and found that the thermal load due to moist air transfer accounts for 75 % of the total cooling load of the system. Their research further revealed that the amount of air transfer significantly depends on the geometrical configuration of the display cases, and also the Reynolds number of the jet at the nozzle exit [20]. Pereria et al. [21] found a 64-fold increase in air transfer in a refrigerator's turned-off fresh food compartment when it was intentionally opened, using the concentration decay method. Liu et al. [22] built a dynamic breathing behavior measurement platform for refrigerator based on the tracer gas technique and estimated that the infiltration suppression effect of magnetic attraction force is approximately $0.061 \text{ g m}^{-1} \text{ N}^{-1}$. They also identified an inhalation effect, the alternating inhalation-exhalation effect and the exhalation, inhalation, exhalation & inhalation effects occur under the pull-down, intermittent refrigeration and defrosting-cooling conditions [23]. The studies mentioned above utilized tracer gas technology to gain a deeper understanding of the moist air transfer in refrigerators under door closing and opening conditions. However, the moist air transfer is influenced by multiple factors, including the operating states under door closing conditions and door opening angle and duration under door opening conditions. Tracer gas technology is also a viable approach for further investigation into these contributing factors.

For refrigeration equipment, the primary pathways for moist air transfer are the gasket, defrosting drainage tube and user door opening behavior. It is essential to ascertain the amount of moisture air transferred and the associated thermal load introduced by each of these pathways. The first two pathways can be mitigated by structural improvements. Liu et al. [22] enhanced the refrigerator gasket structure, reducing the permeability by 5.39 %–62.34 %. Afonso et al. [24] utilized sulfur hexafluoride and adopted the concentration decay method to measure air transfer rates in refrigerators equipped with both new and used gaskets. Their findings revealed that for new gaskets, 3.6 % of the energy used to run the compressor is spent on the air infiltration and 96.4 % on heat gains through the walls. In contrast, the percentages shift to 18.5 % and 81.5 % with used gaskets. Sunghee Kang et al. [25] introduced an innovative defrosting drainage tube design with a hydraulic resistance 220 times greater, resulting in a remarkable 55-fold decrease in inflow as demonstrated through computational fluid dynamics analysis. The action of door opening is inevitable in the usage process of a refrigerator. R. Saidur et al. [26] have noted that ambient air temperature exerts the most substantial influence on energy consumption, with door opening frequency as a secondary factor. They quantified the moist air transfer rate and the resulting energy consumption under both door opening and closing conditions. Their findings indicate energy consumption rose by 8 % due to the operation of door opening-closing as opposed to when the door remained consistently closed. Based on this, they estimated that the additional electricity consumption due to moist air transfer over a 10-year period in Malaysia accounts for 4 % of the total energy

consumption [27]. Alissi et al. [28] found that for a typical door opening schedule, the load for conditions of 91 % relative humidity and a 20 K temperature difference accounted for a 32 % increase in energy usage. Vitor et al. [29] have developed an innovative adaptive defrost control strategy that dynamically adjusts the weighting coefficient of cumulative door opening duration in response to ambient condition variations, such as temperature and humidity, thereby optimizing the timing of predicted defrost initiation. In a simulated test that incorporates door-opening, this adaptive approach achieved a 2.6 % reduction in energy consumption compared to the traditional time-based defrost logic. Generally, the following four categories of loads are associated with door openings: 1) Convective heat transfer, which occurs as ambient air flows over the cooler surfaces of the refrigerator. 2) Latent heat transfer, resulting from the condensation of water vapor present in the moist air that comes into contact with the cooler surfaces of the refrigerator [30]. 3) Radiative heat transfer, which involves heat being transferred from the surrounding environment to the interior surfaces. 4) Sensible heat transfer, which takes place from the external air within the cooled space after the door has been closed [31].

Existing studies have confirmed that the transfer pathways of moist air in refrigerators primarily include the gasket, the defrosting drainage tube, and the door opening behaviors, with the invasion of external moist air adversely affecting the energy consumption of refrigerators [32]. However, there is a lack of quantitative analysis on the transfer rate of moist air through different paths and the introduced thermal load. Particularly, when the refrigerator door is opened, the transfer rate between internal and external moist air significantly increases, leading to a sharp rise in thermal load of the refrigeration system. This can also cause issues such as increase in food temperature, difficulty in reopening the door, and condensation and frosting on the cabinet walls [33,34]. Therefore, to enhance the energy efficiency and performance of refrigerators, an in-depth investigation of the characteristics of moist air transfer in refrigerators under door opening and closing conditions is necessary. This study employs tracer gas technology to measure the transfer rate of moist air through different paths and calculates the introduced thermal load. Specifically, we compared the transfer rate of moist air through the gasket and defrosting drainage tube with the door closed. Also, the transfer rate under different door opening durations and angles with the door opened is compared. Our findings further deepen the understanding of the transfer characteristics of moist air in refrigerators, providing a theoretical basis for improving the energy efficiency and performance of refrigerators.

2. Experimental system and measurement principle

2.1. Description of experiment subject

A refrigerator (Refrigerator A) with a volume of 226 L was selected to investigate the moist air transfer and its thermal load under both door closing and opening conditions. Under door closing conditions, moist air transfer between the interior and exterior of the refrigerator occurred through the gasket and the defrosting drainage tube. And the structural design of the gasket and defrosting drainage tube significantly influences the transfer rate of moist air. In this study, Refrigerator A features a pressure balance hole in its defrosting drainage tube, which can rapidly equalize the pressure inside and outside the refrigerator and also increase the flow of moist air. This study compared Refrigerator A to the freezing compartment of Refrigerator B, which lacks a pressure balance hole in its defrosting drainage tube, to explore the effects of the pressure balance hole. The gaskets and defrosting drainage tube structures of Refrigerator A and Refrigerator B are depicted in Fig. 1. Fig. 2(b) and 2(c) depict the experimental processes for two refrigerators during the measurement of the moist air transfer rate.

In the pull-down process, a drop in temperature and pressure within the refrigerator led to the infiltration of external moist air due to the pressure differential. In the intermittent refrigeration process, the cyclical operation of the compressor induced periodic temperature changes within the refrigerator, leading to oscillations in the pressure differential and a cyclical inhalation and exhalation of moist air, akin to the breathing effect. This study measured the moist air transfer and its thermal load during both the pull-down

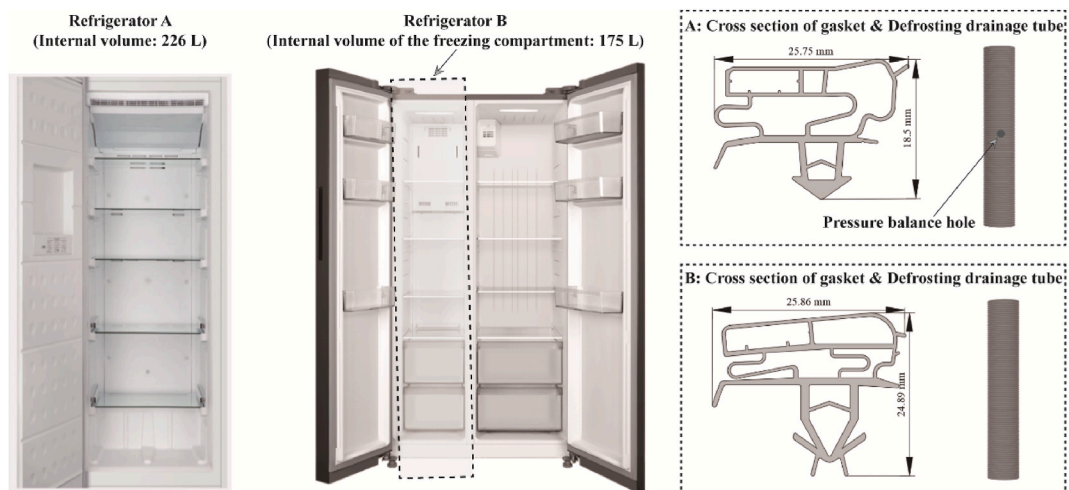


Fig. 1. Two refrigerators for experimental measurements.

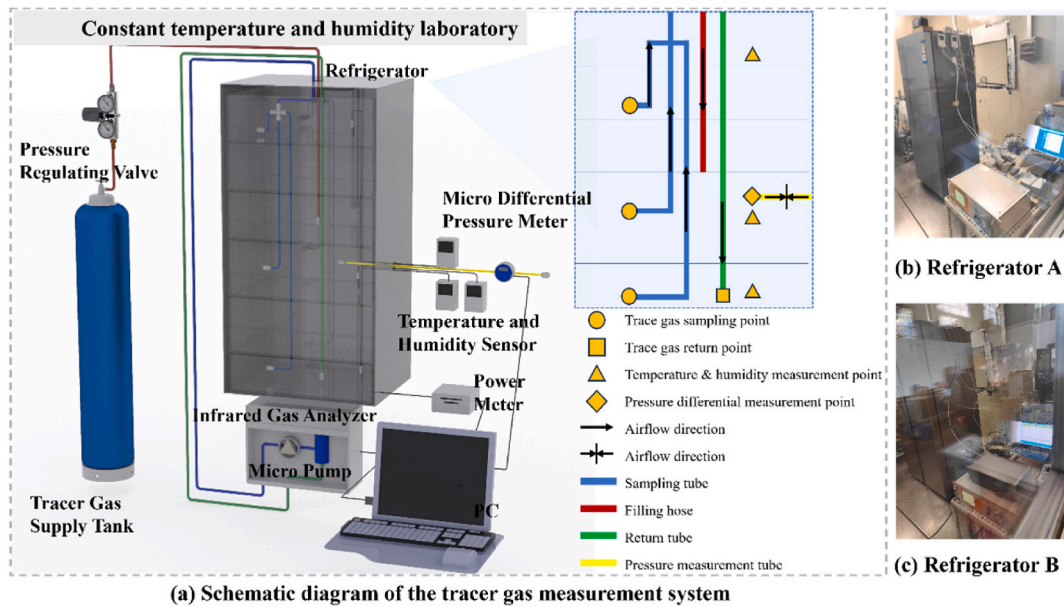


Fig. 2. Image of the moist air transfer rate measurement system.

process and the intermittent refrigeration process under the door closing conditions. Under door opening conditions, intense heat and mass transfer occurred between the internal and external air, which was closely related to the angle and duration of the door opening. Therefore, tests were conducted with door opening angles of 30°, 60° and 90°, and opening durations ranging from 1 s to 5 s. Test conditions for the moist air transfer experiments are listed in Table 1.

2.2. Description of experimental system and testing method

The tracer gas technology was employed to analyze the transfer characteristics of moist air by monitoring changes in the concentration of the tracer gas. The tracer gas selected for this study is carbon dioxide (CO_2), which is commonly used in Heating, Ventilation, and Air Conditioning testing due to its diffusion coefficient being similar to that of moist air and its ability to mix well with air. The typical concentration of CO_2 in ambient air ranges from 300 to 500 ppm, and it is necessary to measure the background CO_2 concentration in the air before the experiment. In this study, the change in CO_2 concentration inside the test refrigerator during a single test does not exceed 1000.0 ppm, and the CO_2 leaked from the refrigerator during the experiment constitutes 0.75 % of the laboratory's ambient CO_2 concentration, which is negligible and does not significantly impact the experimental results.

A tracer gas measurement system, as depicted in Fig. 2(a), was constructed for the study. The system comprises a tracer gas supply tank, a pressure regulating valve, an infrared tracer gas analyzer, temperature and humidity sensors, a micro pressure differential gauge, and the test refrigerator. The details of the instruments utilized in this experimental system are presented in Table 2. To control the external temperature and humidity, the measurement system is situated in a constant temperature and humidity laboratory, with a dry bulb temperature control range of -10.0 to 55.0 °C and a temperature control accuracy of ± 0.2 °C. The relative humidity control range is set from 35.0 % to 85.0 %, with a humidity control accuracy of ± 2.5 %.

During the measurement, a micro pump integrated with the infrared gas analyzer continuously extracted gas from the sampling points. After the measurement of the sampling tracer gas concentration by the analyzer, the sampling gas was recirculated back into the refrigerator via the return tube. Sampling points were strategically placed at the top, middle, and bottom of the refrigerator to monitor the uniform distribution of tracer gas. Additionally, a lower sampling gas flow rate of 50.0 mL min^{-1} was employed to minimize the disturbance of the flow field caused by the sampling gas.

Table 1
Test conditions for the moist air transfer experiments.

Case	Test box	Environmental conditions	Setting temperature	Experimental purposes
Case 1	Refrigerator A and B	25 °C&50 %	-18 °C	comparing air transfer path
Case 2	Refrigerator A	25 °C&69.3 %	-24 °C	obtaining air transfer rate in pull-down process
Case 3	Refrigerator A	25 °C&69.3 %	-24 °C	obtaining air transfer rate in intermittent refrigeration process
Case 4	Refrigerator A	25 °C&69.3 %	-24 °C	obtaining air transfer rate in door opening-closing process

Table 2
Specifications of instrumentation.

Instrumentation	Manufacturer	Model	Full scale	Accuracy
Thermocouple	Omega	Type T	−50~200 °C	±0.5 °C
Humidity sensor	Elitech	GSP-6	10~99 % RH	±5 %
Differential pressure meter	Dwyer	MSX	−25~25 Pa	±1 %
Infrared gas analyzer	Gechee	IRME-G	0~1500 ppm	±1 %

2.3. Mass balance equations and solution methods

Assume that the moist air transfer between the measured refrigerator and external environment is in a steady state and the tracer gas is uniformly distributed inside the refrigerator. Then, the transient mass balance equation of the tracer gas inside and outside the refrigerator can be expressed as Eq. (1) [24]:

$$\nu \times dt - (c - c_{\text{atm}}) \times \dot{V} \times dt = \left(\frac{dc}{dt} \right) \times dt \times V \quad (1)$$

Where ν is the volumetric charging rate of tracer gas, $\text{m}^3 \cdot \text{h}^{-1}$. V is the internal volume of the test refrigerator, m^3 . c is the real time tracer gas concentration inside the measured refrigerator, ppm. c_{atm} is the background tracer gas concentration of the external environment, ppm. $\dot{V}$ is the air volume transfer rate between the measured refrigerator and external environment, $\text{m}^3 \cdot \text{h}^{-1}$.

To solve Eq. (1), two strategies can be employed: the straight-line method and the integral method. The straight-line method estimates the total transfer of moist air over a period by determining the average decay rate of CO_2 based on its concentration at the initial and final moments. In contrast, the integral method integrates the decay rates of CO_2 at each moment to accumulate the total transfer of moist air [24,35]. When using the straight-line method, the transfer ratio (I) of internal and external air is initially defined as:

$$I = \frac{\dot{V}}{V} \quad (2)$$

Subsequently, Eq. (1) can be deformed to the first-order nonlinear equation, as expressed in Eq. (3):

$$\frac{dc}{dt} + I \times c = I \times c_{\text{atm}} + \frac{\nu}{V} \quad (3)$$

In the concentration decay method, the tracer gas is pre-charged into the refrigerator. Hence, the volumetric charging rate (ν) of tracer gas is set to 0. With the measured initial concentration of the tracer gas (c_0) in measured refrigerator specified, the solution of Eq. (3) can be derived as Eq. (4):

$$c = (c_0 - c_{\text{atm}}) \times e^{-I \times t} + c_{\text{atm}} \quad (4)$$

Continue to deform Eq. (4), the concentration decay equation of the tracer gas can be obtained:

$$\ln(c - c_{\text{atm}}) = -I \times t + \ln(c_0 - c_{\text{atm}}) \quad (5)$$

In Eq. (5), $\ln(c - c_{\text{atm}})$ decays with time at a rate characterized by the slope (I) and $\ln(c_0 - c_{\text{atm}})$ is known constant. Finally, the moist air transfer ratio (I) and the air volume transfer rate between the interior and exterior of the refrigerator can be determined.

When using the integral method, the mass balance equation for the tracer gas after t_1 can be obtained as Eq. (6) by integrating Eq. (1) in time.

$$\nu \times t_1 + c_{\text{atm}} \times \dot{V} \times t_1 - \dot{V} \times \int_0^{t_1} c \, dt = V \times (c_{t_1} - c_0) \quad (6)$$

where c_{t_1} is the tracer gas concentration in the measured refrigerator after t_1 , ppm.

As mentioned above, when the concentration decay method is used, ν is 0. Under steady-state conditions, $\dot{V}$ is constant and can be expressed as Eq. (7):

$$\dot{V} = \frac{V \times (c_{t_1} - c_0)}{c_{\text{atm}} \times t_1 - \int_0^{t_1} c \, dt} \quad (7)$$

Once the volumetric transfer rate $\dot{V}$ is obtained, we can calculate the mass transfer rate $\dot{m}$:

$$\dot{m} = \rho \times \dot{V} \quad (8)$$

Where ρ is the density of air at the average air temperature inside and outside the refrigerator, $\text{kg} \cdot \text{m}^{-3}$.

2.4. Uncertainty analysis

The uncertainty analysis was performed following the guidance provided by Robert J [36]. In this experiment, the tested parameters can be divided into direct measurement parameters and indirect measurement parameters. Direct measurement parameters such as temperature, relative humidity and pressure difference are directly obtained from the measuring instruments. Therefore, their uncertainty is determined by the accuracy of the sensor. The specifications of the measuring instruments are listed in Table 2. The uncertainty of indirect parameters is calculated by following the uncertainty propagation rule and can be expressed in Eqs. (9) and (10). The uncertainty analysis of indirect parameters is shown in Table 3.

$$R = (X_1, X_2, X_3 \dots X_N) \quad (9)$$

$$U_R = \frac{\delta R}{R} = \frac{\sqrt{\left(\frac{\partial R}{\partial X_1} X_1\right)^2 + \left(\frac{\partial R}{\partial X_2} X_2\right)^2 + \dots + \left(\frac{\partial R}{\partial X_N} X_N\right)^2}}{R} \quad (10)$$

3. Comparison of solving methods and moist air transfer paths under door closing conditions

3.1. Experimental measurement process

The experimental works were implemented in a constant temperature and humidity laboratory. The environmental temperature was set as 25.0 °C and relative humidity was set as 50 %. During the experiment, the temperatures of Refrigerator A and B were set at −18 °C and Fig. 3 shows the temperature and relative humidity variations in two refrigerators. However, Fig. 3 reveals that despite a constant target temperature, the actual temperatures were not completely stable. Specifically, the temperature of Refrigerator A ranged from −19.5 °C and −16.5 °C, while the temperature of Refrigerator B temperature ranged from −19.1 °C to −17.1 °C. These fluctuations are attributed to the range-based temperature control strategy of refrigerators. When the temperature at the control point drops below the lower limit, the compressor and fan are off, causing the temperature to gradually increase. Conversely, when the temperature exceeds the upper limit, the compressor and fan are on, causing the temperature to decrease, thus creating a cycle for temperature regulation. Furthermore, the relative humidity of the air also exhibited cyclical fluctuations, closely followed the temperature trends. The reason is that as the temperature decreases, the water vapor content of moist air diminishes, consequently lowering the relative humidity. The fluctuation ranges of the relative humidity inside Refrigerator A and Refrigerator B were 55.3 %–68.1 % and 37.6 %–60.6 %. As depicted in the figure, the temperature fluctuation period of Refrigerator A is significantly longer than that of Refrigerator B, lasting 0.88 h versus 0.52 h. This is because Refrigerator A is a low-temperature freezer, which can reach a cooling system temperature of −40 °C. The enhanced insulation of Refrigerator A leads to a more gradual internal temperature increase when the fan and compressor are off, consequently prolonging its operational cycle.

Under the door closing conditions, the main air transfer paths of the refrigerator are gasket and defrosting drainage tube. Fig. 4(a) ~ Fig. 4(b) show the variation trend of CO₂ concentration in various working phases such as ambient, filling, pull-down and intermittent refrigeration. Since the CO₂ concentration of Refrigerator A decreases rapidly without sealing the defrosting drainage tube, it is necessary to carry out two or three times of gas filling during the measurement process. The background concentration of CO₂ in the environment is stable, and the value of this concentration is the c_{atm} in Eq. (5) and Eq. (7). As shown in Fig. 4(a) and (b), when the collection time was more than 1 h, the fluctuation of background concentration of CO₂ was less than 3.0 ppm. In the initial filling stage, due to the uneven distribution of CO₂ concentration, the concentration collected by the infrared gas analyzer reached the maximum range of 1500.0 ppm. After 10 min, it was evenly distributed in the refrigerator. The concentration decreased to a stable value and no longer changed. At this moment the concentration is c_0 in Eq. (5) and Eq. (7). According to Eq. (5), Fig. 5(a) ~ Fig. 5(f) give the linear fitting results of the CO₂ concentration of two refrigerators in the door closing conditions using straight-line method.

3.2. Comparison of solving methods

According to Eq. (5), Eq. (7) and Eq. (8), the moist air volume transfer rate ($\dot{V}$, m³·h^{−1}) and mass transfer rate ($\dot{m}$, g·h^{−1}) of Refrigerator A and Refrigerator B were solved by the straight-line method and integral method, as shown in Table 4.

It can be seen from Table 4 that the deviation of moist air transfer rate under the two solving methods is within ±4 %. Since the integral method needs to perform discrete integral calculation on the CO₂ concentration of each acquisition cycle, the solving process is cumbersome. Thus, the straight-line method is used to solve $\dot{m}$. Fig. 5(a)–Fig. 5(f) show the fitting results obtained by straight-line

Table 3
Relative uncertainties of derived parameters.

Parameters	Relative uncertainty
Mass transfer rate ($\dot{m}$)	3.49 %
Sensible thermal load (Q_{sen})	7.28 %
Latent thermal load (Q_{lat})	8.83 %

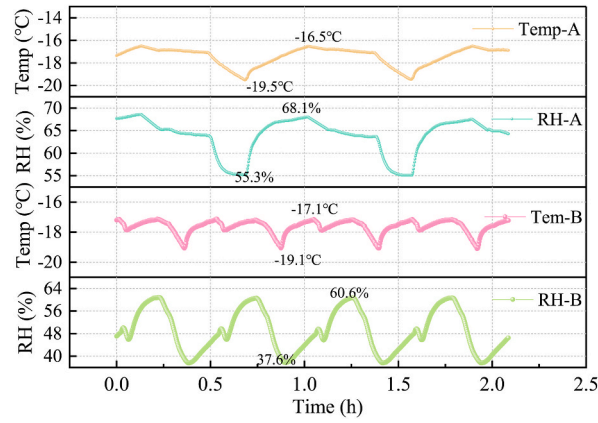
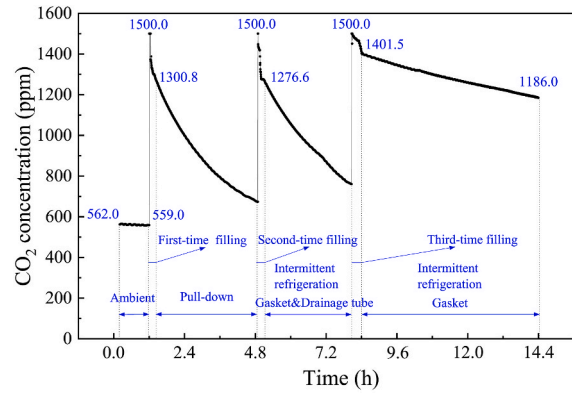
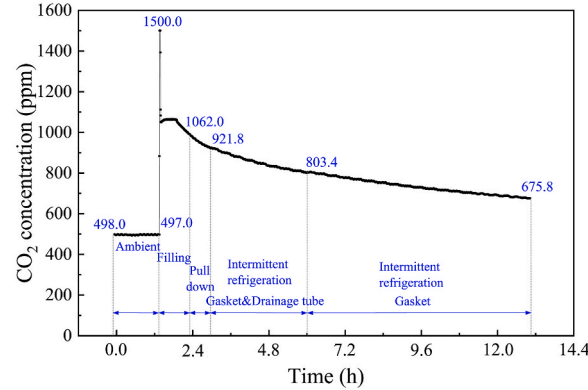


Fig. 3. Temperature and relative humidity variations in two refrigerators.



(a) Refrigerator A



(b) Refrigerator B

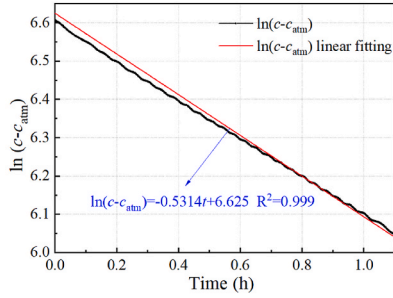
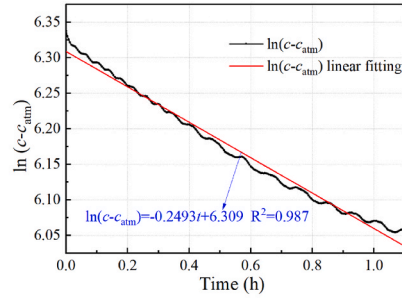
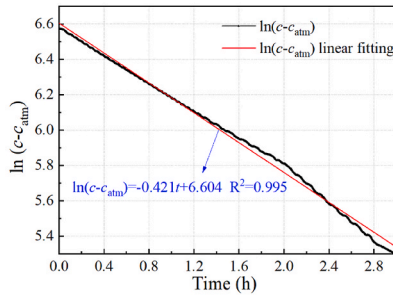
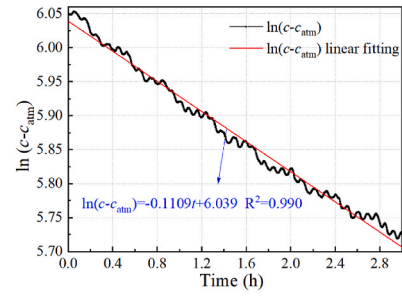
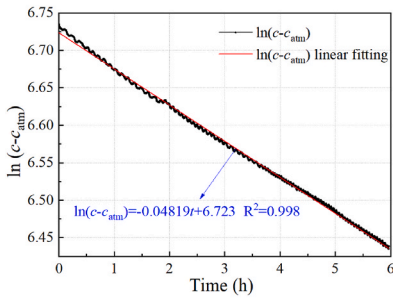
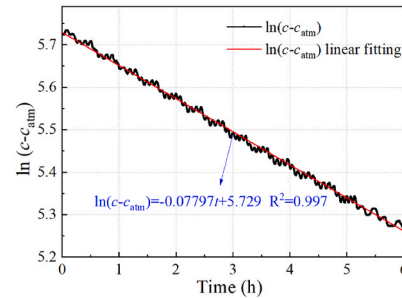
Fig. 4. Curves of CO₂ concentration at the measurement process.

method, and it can be seen that the deciding coefficient ($R^2 = 1 - \frac{\sum (y_{\text{actual}} - y_{\text{predict}})^2}{\sum (y_{\text{actual}} - y_{\text{mean}})^2}$) of the fitted line is 0.987–0.999. This shows that the relationship between the fitting correlation and the measured value is strong. As shown in Fig. 5(a), the slope of the fitted curve (I) is 0.5314 h^{-1} . It means that the transfer ratio of internal and external air in Refrigerator A is 53.14 % of the cabinet volume in 1 h, i.e. 120.0 L.

Table 4

Calculated results of moist air transfer rate under two solving methods.

Refrigerator	Conditions	Straight-line method		Integral method		Deviation
		$\dot{V}$ ($10^{-2} \text{ m}^3 \text{ h}^{-1}$)	$\dot{m}$ ($\text{g} \cdot \text{h}^{-1}$)	$\dot{V}$ ($10^{-2} \text{ m}^3 \text{ h}^{-1}$)	$\dot{m}$ ($\text{g} \cdot \text{h}^{-1}$)	
Refrigerator A	Pull-down (Unsealed drainage tube)	12.000	120.000	12.03	120.30	0.25 %
	Intermittent refrigeration (Unsealed drainage tube)	9.514	95.140	9.511	95.110	-0.03 %
	Intermittent refrigeration (Sealed drainage tube)	1.085	10.850	1.097	10.970	1.11 %
Refrigerator B	Pull-down (Unsealed drainage tube)	4.375	43.750	4.393	43.925	0.40 %
	Intermittent refrigeration (Unsealed drainage tube)	1.998	19.979	1.925	19.250	-3.65 %
	Intermittent refrigeration (Sealed drainage tube)	1.361	13.611	1.356	13.563	-0.36 %

(a) Pull-down process of Refrigerator A
(Unsealed drainage tube)(b) Pull-down process of Refrigerator B
(Unsealed drainage tube)(c) Intermittent refrigeration process of
Refrigerator A
(Unsealed drainage tube)(d) Intermittent refrigeration process of
Refrigerator B
(Unsealed drainage tube)(e) Intermittent refrigeration process of
Refrigerator A (Sealed drainage tube)(f) Intermittent refrigeration process of
Refrigerator B (Sealed drainage tube)**Fig. 5.** Curves of $\ln(c - c_{\text{atm}})$ and its linear fitting.

3.3. Transfer path analysis

The main air transfer paths of a refrigerator under door closing conditions mainly consist of the gasket and defrosting drainage tube. Therefore, the combined transfer rate of moist air at the gasket and defrosting drainage tube represents the total rate of moist air transfer for the refrigerator under the door closing conditions. When the defrosting drainage tube is deliberately sealed, the measured transfer rate corresponds to that through the gasket alone. By calculating the difference between the total transfer rate at both the gasket and defrosting drainage tube and the rate at the gasket, the transfer rate through the defrosting drainage tube can be determined. Based on Tables 4 and In the intermittent refrigeration process, when the defrosting drainage tube is unsealed, the total mass transfer rates of moist air in Refrigerator A and Refrigerator B are 95.110 g h^{-1} and 19.250 g h^{-1} . When the defrosting drainage tube is sealed, the mass transfer rates of moist air through gasket in Refrigerator A and Refrigerator B are 10.970 g h^{-1} and 13.563 g h^{-1} . Then, the transfer proportion of gasket and defrosting drainage tube in two refrigerators can be calculated, as shown in Fig. 6. It can be seen that the difference in the proportion occupied by the defrosting drainage tube was significant, reaching 88.5 % with a balance hole and 29.5 % without a balance hole. This suggests that when the defrosting drainage tube is equipped with a pressure balance hole, it will significantly increase the transfer rate of moist air.

4. Moist air transfer characteristics under door closing conditions

The experimental works were implemented in a constant temperature and humidity laboratory. According to the test method mentioned in Household and similar refrigerating appliances (GB/T 8059-2016), the environmental temperature was set as $25.0 \text{ }^{\circ}\text{C}$ and relative humidity was set as 69.3 %. Refrigerator A was utilized for further experimental investigations on moist air transfer characteristics under door closing and opening conditions.

4.1. Pull-down condition

Fig. 7(a) shows curves of operating parameters in pull-down process, including temperature, relative humidity, pressure difference (PD), and power (P). After the compressor started-up, the temperature inside the cabinet dropped from $24.5 \text{ }^{\circ}\text{C}$ to $-26.6 \text{ }^{\circ}\text{C}$. Subsequently, the compressor stopped and transit into the intermittent refrigeration process. The pull-down period was about 2.35 h with the cooling rate of $21.8 \text{ }^{\circ}\text{C h}^{-1}$. Different from the temperature change trend, the relative humidity in the cabinet increased from 80.8 % to 87.3 % at about 1 min after compressor started-up. In this stage, the air moisture content in the cabinet kept constant, and the increase of relative humidity was caused by the slight decrease of temperature. Subsequently, the temperature decreases, resulting in a decrease in the saturation pressure of the water vapor. Meanwhile, with the infiltration of external high moist air through the gasket and defrosting drainage tube, relative humidity should have risen. However, the water vapor condensed on the evaporator surface, causing a decrease in the moisture content, and the relative humidity was finally reduced to 28.1 %.

According to the ideal gas equation of state $pV = nRT$, when the temperature decreased, the pressure in the cabinet also decreased, that is, the cabinet was in a negative pressure state. The start-up of the compressor caused a remarkable decrease in the local temperature, which brought a pulsating negative pressure of -4.1 Pa . In our previous study [22], this pulsating negative pressure was -13.1 Pa when sealing the defrosting drainage tube. It infers that this pressure is related to the sealing strength of cabinet. The external moist air was inhaled into the cabinet through the gasket and defrosting drainage tube. The large negative pressure formed by the

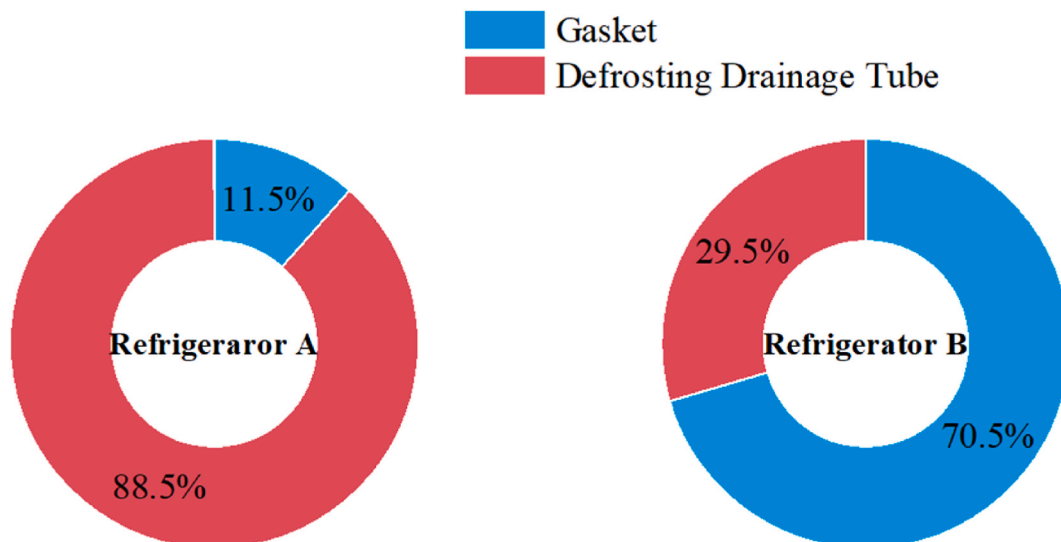


Fig. 6. Transfer path of refrigerator under door closing conditions.

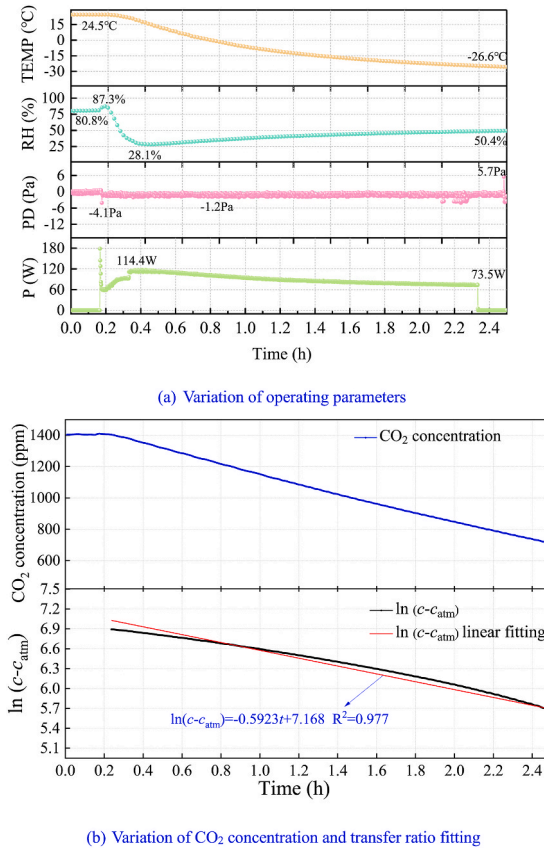


Fig. 7. Variation of operating parameters and CO₂ concentration in pull-down process.

startup of the compressor was gradually balanced by inhaled moist air and finally maintained at about -1.2 Pa. As the relative humidity dropped at 28.1 %, the condensation of water vapor on the evaporator surface led to the decrease of moisture content. Thereafter the external moist air was constantly inhaled into the cabinet to supply water vapor and the temperature dropping also caused the water vapor saturation pressure to decrease. Therefore, the relative humidity began to increase and reached 50.4 % at the end of the pull-down process. As shown in Fig. 7(b), as the moist air with low-concentration CO₂ in the external environment was inhaled into the cabinet in the pull-down process, the CO₂ concentration inside the cabinet was diluted. The fitted I was 0.592 h^{-1} and $\dot{m}$ was 133.018 g h^{-1} .

4.2. Intermittent refrigeration condition

From Fig. 8(a), it can be seen that the temperature inside the cabinet fluctuated at $-26.5^\circ\text{C} \sim -22.5^\circ\text{C}$ with the on-off of the compressor, while the relative humidity fluctuated at 44.2 %–58.5 %. Once the temperature was higher than the upper limit value, compressor started-up and the pressure dropped sharply at the moment of compressor started-up, forming a pulsating negative pressure of about -9.1 Pa. The external moist air was constantly inhaled into the cabinet, the negative pressure gradually dropped. Once the temperature was lower than the lower limit value, compressor stopped and the pressure rose sharply at the moment of compressor shutdown, forming a pulsating positive pressure of about 6.1 Pa. Internal moist air was exhaled outside the cabinet through the gasket and defrosting drainage tube, the positive pressure also gradually dropped. Fig. 8(b) shows the variation curve of CO₂ concentration and its transfer ratio fitting.

It can be seen from Fig. 8(b) that the CO₂ concentration in the cabinet was diluted during the inhalation and exhalation process. Each inhalation-exhalation behavior was performed, external moist air with low CO₂ concentration replaced internal moist air with high CO₂ concentration. The fitted transfer ratio I in the intermittent process was 0.455 h^{-1} and $\dot{m}$ was 102.920 g h^{-1} .

5. Moist air transfer characteristics under door opening-closing conditions

The experimental works in this section were also carried out at the constant ambient temperature and humidity with 25.0°C and 69.3 %. The door opening angle of the refrigerator was realized by preset a stroke limiter. The door opening-closing duration was realized by manual punching a time clock and the error was within ± 15.0 %.

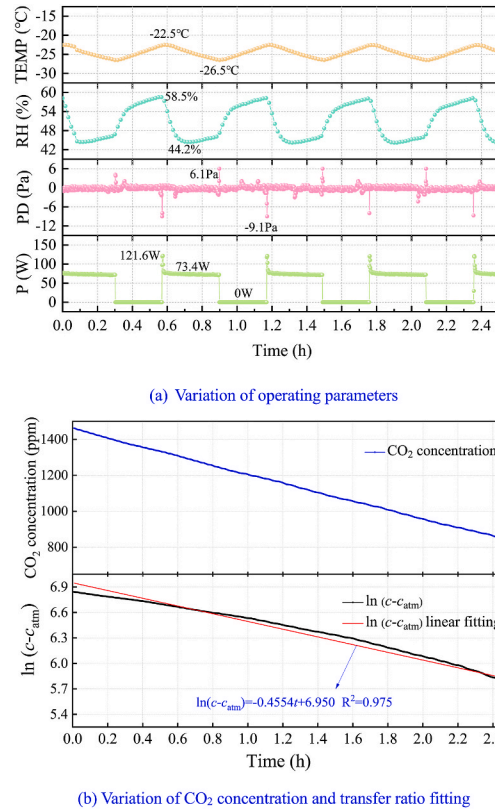


Fig. 8. Variation of operating parameters and CO₂ concentration in intermittent refrigeration process.

5.1. Operating parameter variation characteristics

Fig. 9(a) shows the variation curves of moist air parameters at a 30° door opening angle for different door opening-closing durations. The longer the door opening-closing duration, the greater the fluctuation in temperature, relative humidity and pressure difference. As the duration increased from 1 s to 5 s, the temperature peaks raised at −24.6 °C, −23.5 °C, −22.9 °C, −20.3 °C and −19.7 °C and the relative humidity peaks raised at 73.1 %, 78.0 %, 84.8 %, 92.8 % and 95.6 %. It can be inferred that the inhalation of external moist air and the exhalation of internal moist air occurred simultaneously under the door opening-closing conditions. Meanwhile, the pressure difference valley points dropped at −100.0 Pa, −550.3 Pa, −1043.8 Pa, −1143.2 Pa and −1425.1 Pa. A greater transfer of moist air resulted in a more significant drop in temperature and relative humidity after the door closing, leading to a more remarkable negative pressure and secondary door opening force. The magnitude of the negative pressure was directly proportional to the contact stress on the sealing surface. As the negative pressure value decreased, it reduced the inhalation of moist air and prolonged the time required to re-establish pressure equilibrium. Based on the maximum opening force of 80 N, the threshold negative pressure for door operability was determined to be −31.2 Pa, with the secondary door opening time defined as the duration until the negative pressure was balanced to higher than −31.2 Pa. For example, the maximum door opening force and secondary door opening time at a 1 s door opening-closing duration was 136.6 N and 25 s. The maximum door opening force and the secondary opening time at a 5 s duration was as high as 1221.9 N and 55 s, showing an increase of 8.9 times and 2.2 times.

Regarding the initial CO₂ concentration inside the cabinet as the background value, the effect of door opening-closing durations on the transfer rate (I) of each second was examined, as depicted in Fig. 9(b). In Fig. 9(b), I_{51} denotes the rate of air exchange occurring within the first second of the test, under conditions of a 5-s door opening and closing duration. During the door opening-closing process, there was a sharp decrease in the internal CO₂ concentration of the cabinet. As the door opening-closing duration increased from 1 s to 5 s, the transfer rate exhibited a progressive decline. When the duration exceeded 2 s, the variation in I tended to stabilize from the 2 nd s. This trend was due to the negative pressure within the cabinet prior to door opening, which led to a rapidly air transfer after door opening. The rapid equalization of this pressure difference led to a gradual decrease in I . During door closing process, the reduction in air flow area resulted in a further decrease in I . The closing of the door may also cause the displaced air to flow back into the cabinet, leading to negative values of I . Under conditions of a 30° door opening angle and a 5 s door opening-closing duration, the transfer rate was calculated to be 1.450 s^{-1} (I_{51}) at the first second, diminishing to 0.010 s^{-1} (I_{55}) by the fifth second.

When the door opening angle increased to 60°, the moist air transfer area was enlarged, intensifying the disturbance to the cabinet temperature and humidity. As shown in Fig. 10(a), the internal temperature of the cabinet rose over −21.0 °C and the relative humidity surpassed 80.0 % under different opening-closing durations. The time required to return to normal refrigeration after door closing was

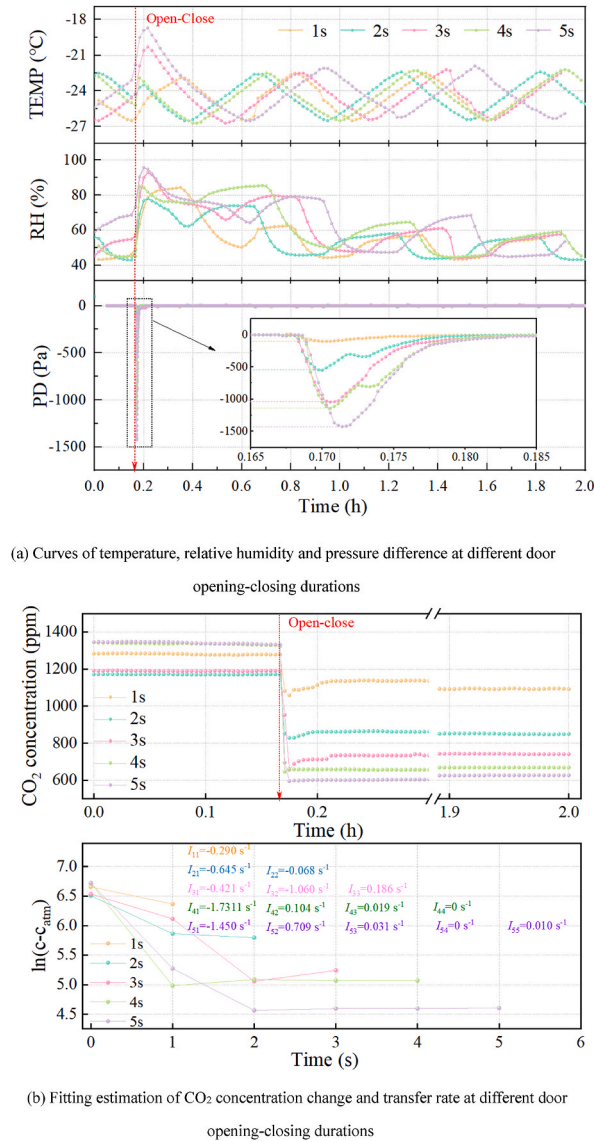


Fig. 9. Experiment of moist air transfer at 30° door opening angle.

longer. Compared to the 30° door opening angle, the disturbance to the relative humidity remains greater than that to temperature, and the pressure difference valley point within the cabinet after door closing was even lower. For instance, for a 1 s door opening-closing duration, the pressure difference valley point was -518.7 Pa, which was about five times lower than when the door was opened at 30° angle. Particularly for a 5 s door opening-closing duration, the pressure difference valley point dropped to -1800.3 Pa, generating a significant sealing force on the door. In Fig. 10(b), the CO₂ concentration also decreased rapidly at 60° than that at 30°. The transfer rate (I) showed a consistent downward trend. Under conditions of a 60° door opening angle and a 5 s door opening-closing duration, the transfer rate I was calculated to be 1.457 s^{-1} (I_{51}) at the first second, diminishing to 0.013 s^{-1} (I_{55}) by the fifth second.

90° is the maximum opening angle of the refrigerator door. As shown in Fig. 11(a), the temperature and humidity increased to a greater extent as the door opening-closing duration increased. For a 5 s door opening-closing duration, the temperature and humidity rose to -19.1 °C and 98.0 %. With the door opening-closing duration increased from 1 s to 5 s, the pressure difference valley point reached at -706.2 Pa, -1218.7 Pa, -1368.8 Pa, -1525.5 Pa and -1856.8 Pa. It led to a large deformation the door gasket and the flexible rubber strip became compressed. Due to the large contact stress of the gasket, the transfer gap for the moist air was minimal, causing a decrease in moist air transfer rate. Consequently, compared with 30° door opening angle, the time required for negative pressure equilibrium was significantly prolonged. In Fig. 11(b), under conditions of a 90° door opening angle and a 5 s door opening-closing duration, the transfer rate I was calculated to be 2.262 s^{-1} (I_{51}) at the first second, diminishing to 0.20 s^{-1} (I_{55}) by the fifth second.

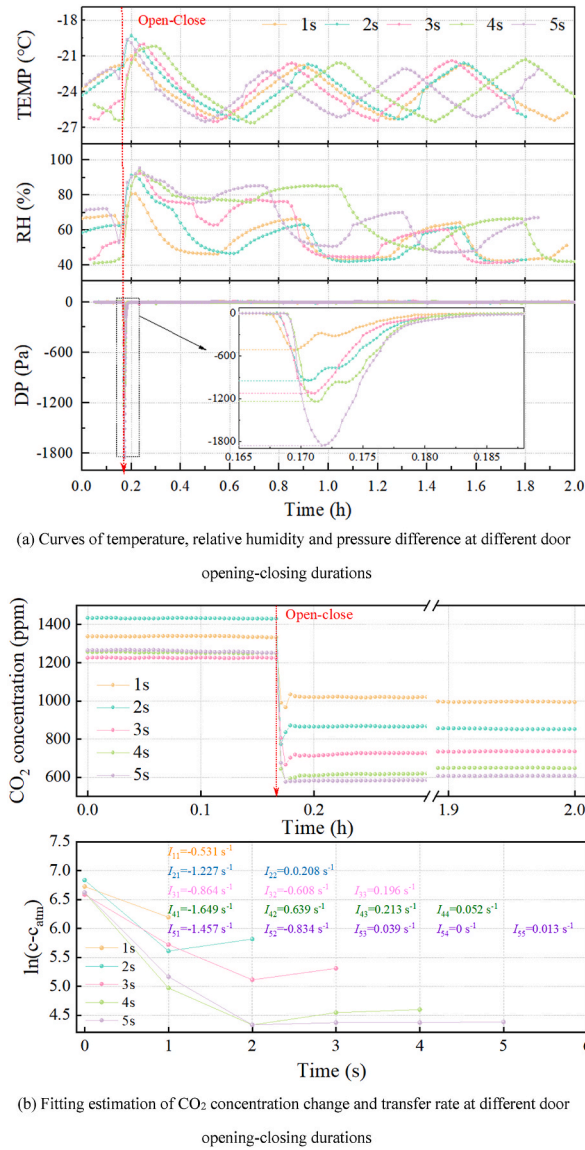
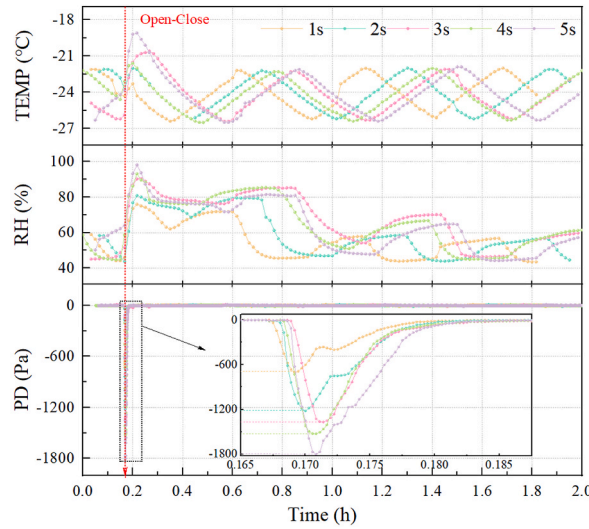


Fig. 10. Experiment of moist air transfer at 60° door opening angle.

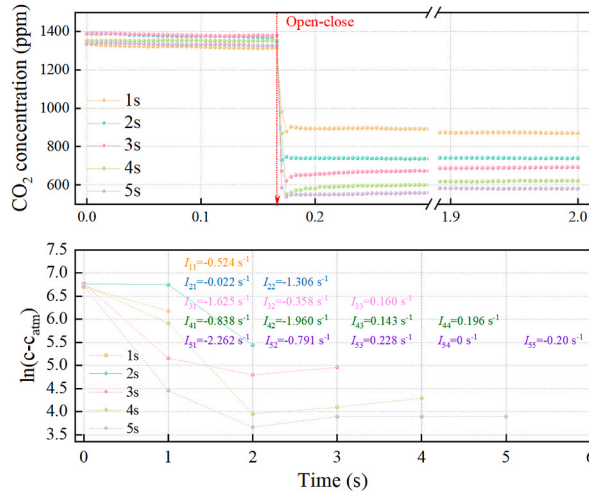
5.2. Influence of door opening angle and opening-closing duration

Under varying conditions of door opening angles and durations, the mass transfer rate $\dot{m}$ was as high as 65.598 g s^{-1} – 150.076 g s^{-1} , which was approximately 2300–4500 times higher than in the door closing scenario. The effect of door opening angle and duration on $\dot{m}$ is depicted in Fig. 12. An upward trend in mass transfer is observed with increasing angles and durations. A larger door opening angle results in a greater area for moist air transfer, thereby increasing the volume of moist air transferred. For angles of 30°, 60° and 90°, the leakage areas are approximately 0.576 m^2 , 1.082 m^2 and 1.463 m^2 . As the door opening angle increases, the mass transfer per unit area continuously decreases. This is because the larger the leakage area, the faster the pressure differential between the inside and outside of the cabinet reaches equilibrium, reducing the drive for moist air transfer. Similarly, with longer door opening durations, the greater the decrease in CO₂ concentration, and correspondingly, the higher the mass transfer rate.

The trends in maximum door opening force and secondary door opening time were similar, as depicted in Fig. 13. As mentioned earlier, with increased door opening angles and durations, the mass transfer rate of moist air increased, leading to a greater negative pressure during the cooling and dehumidification process inside the cabinet. The formation of this negative pressure was similar to that in pull-down process, i.e. the rapid decrease of temperature and humidity of moist air near the evaporator caused the sudden drop of local pressure. Overall, as the door opening angle increased from 30° to 60° and then 60°–90°, the average maximum door opening force increased by 29.83 % and 17.63 %. The average secondary door opening time extended by 21.00 % and 8.68 %. The range of



(a) Curves of temperature, relative humidity and pressure difference at different door opening-closing duration



(b) Fitting estimation of CO₂ concentration change and transfer rate at different door opening-closing duration

Fig. 11. Experiment of moist air transfer at 90° door opening angle.

secondary door opening times for the three different door angles were 25 s–55 s, 43 s–62 s and 47 s–67 s. The greater the negative pressure, the greater the inward sealing force formed, which slowed the inhale rate of external moist air and prolonged the time for pressure equilibrium. To address the issue of difficulty in secondary door opening, this study recommends the installation of a pressure relief valve on the refrigerator door, which can be activated to rapidly balance the negative pressure inside the cabinet when a secondary door opening is required within a short time frame.

6. Comparison of moist air transfer thermal load under door closing and opening conditions

6.1. Calculation method for thermal load

When external moist air transfers into the cabinet, the dry air component is cooled down to the same temperature as the interior air, thereby generating the sensible thermal load (Q_{sen}), which can be determined using Eq. (11):

$$Q_{\text{sen}} = \frac{\dot{m}}{3.6 \times 10^3} \frac{\rho_{\text{dry}}}{\rho} \bar{c}_{p,a} (T_{\text{out}} - T_{\text{in}}) \quad (11)$$

where $\dot{m}$ is the mass transfer rate, $\text{g} \cdot \text{h}^{-1}$. ρ_{dry} is the density of dry air, $\text{kg} \cdot \text{m}^{-3}$. ρ is the density of moist air, $\text{kg} \cdot \text{m}^{-3}$. $\bar{c}_{p,a}$ is the average

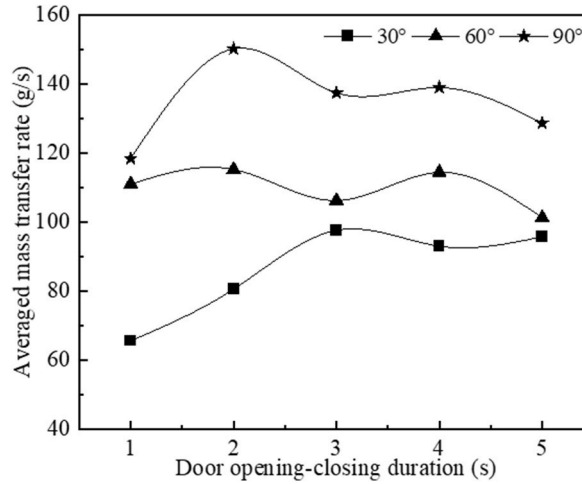


Fig. 12. The variation of $\dot{m}$ with different door opening angles and durations.

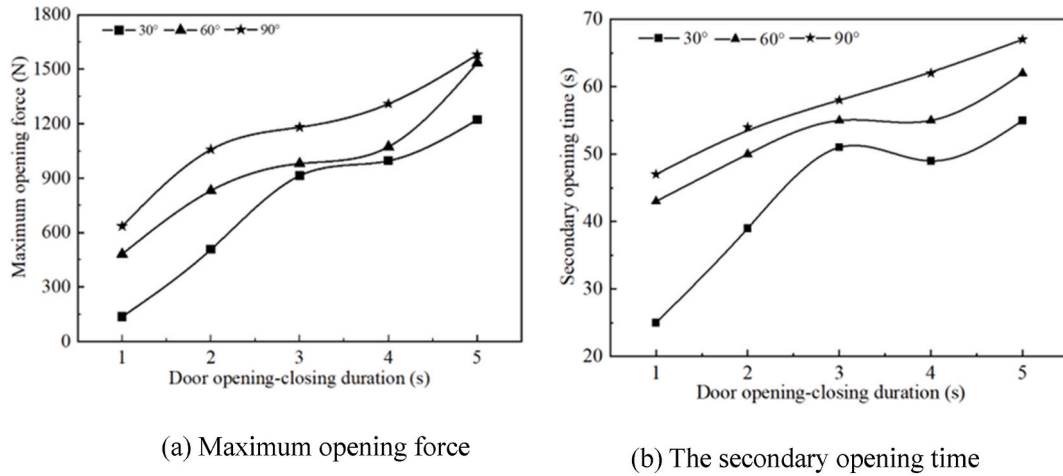


Fig. 13. The influence of the door opening angle and durations on the maximum opening force and the secondary opening time.

specific heat capacity at constant pressure for dry air as it cools from temperature T_{out} to T_{in} , $\text{kJ} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$. T_{in} and T_{out} are the inner and outer temperatures of the cabinet, $^{\circ}\text{C}$.

The water vapor in moist air undergoes three distinct processes upon entering into the cabinet: 1) A portion of the water vapor is cooled down to the same temperature as the interior air and continues to coexist with dry air in the gaseous state. 2) Another part of the water vapor condenses into liquid on the walls, sometimes further solidifying into ice. 3) The majority of the water vapor sublimates on the evaporator surface to form a frost layer. The latter two processes, involving phase transitions of water vapor, are responsible for the latent thermal load, which is commonly assessed by the disparity in moisture content between the ambient and internal air. For computational simplicity, the sublimation process is often neglected, with the assumption that water vapor experiences cooling, condensation, and frosting. Consequently, the sensible thermal load (Q_{lat}) is determined through Eq. (12) and Eq. (13).

$$m_d = \int 10^{-3} \cdot \dot{m} \frac{\rho_{dry}}{\rho} \times (d_{out} - d_{in}) dt \quad (12)$$

$$Q_{lat} = \frac{m_d}{3.6 \times 10^3 t} (\bar{c}_{p1} T_{out} + r_{wat} + r_{ice} + \bar{c}_{p2} T_{eva}) \quad (13)$$

where m_d is the mass of water vapor transfer into the cabinet, g . t is the duration of refrigeration operation, h . $\bar{c}_{p1}$ is the average specific heat capacity at constant pressure for water vapor as it cools from T_{out} to 0°C , approximately $1.850 \text{ kJ} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$. r_{wat} is the latent heat of vaporization of water, approximately $2257.6 \text{ kJ} \cdot \text{kg}^{-1}$. r_{ice} is the latent heat of fusion for ice, approximately $336 \text{ kJ} \cdot \text{kg}^{-1}$. $\bar{c}_{p2}$ is the average specific heat capacity at constant pressure for ice as it cools from 0°C to T_{eva} , approximately $2.100 \text{ kJ} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$. T_{eva} is the average temperature of the evaporator surface during refrigerator operation, measured by a thermocouple, at -30°C . d_{in} and d_{out} are

the moisture content of internal cabinet air and external environment air, $\text{g} \cdot (\text{kg}_{\text{dry air}})^{-1}$.

Consequently, the thermal load (Q) caused by moist air transfer is expressed as Eq. (14):

$$Q = Q_{\text{sen}} + Q_{\text{lat}} \quad (14)$$

When considering the operational durations (t), the accumulative thermal load, i.e., cooling consumption (E) caused by heat and moisture transfer loads can also be determined as Eq. (15):

$$E = E_{\text{sen}} + E_{\text{lat}} \quad (15)$$

In addition to the thermal load introduced by moist air transfer, the total thermal load of the refrigerator also includes the thermal loads of the cabinet walls and gasket. These thermal loads can be calculated using Eqs. (16) and (17).

$$Q_{\text{wall}} = \frac{A(T_{\text{out}} - T_{\text{in}})}{\frac{1}{h_{\text{wall,out}}} + \sum_i^n \frac{\delta_i}{\lambda_i} + \frac{1}{h_{\text{wall,in}}}} \times t \quad (16)$$

$$Q_{\text{gasket}} = h_{\text{gasket}} L (T_{\text{out}} - T_{\text{in}}) \times t \quad (17)$$

where Q_{wall} is the thermal load of the cabinet walls, kJ. $h_{\text{wall,out}}$ and $h_{\text{wall,in}}$ are convective heat transfer coefficient for the outer and inner walls of the refrigerator, with values taken as $10 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ and $20 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$, respectively [37]. A is the area of the wall, m^2 . δ_i is the thickness of the wall, mm. λ_i is the thermal conductivity of different materials, $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. Q_{gasket} is the thermal load of the gasket, kJ. h_{gasket} is the heat transfer coefficient per unit length of the gasket, with value taken as $0.04619 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ [22]. L is the length of the gasket, m.

6.2. Calculation results for thermal load

The moisture content of ambient air was measured at approximately $13.790 \text{ g} \cdot (\text{kg}_{\text{dry air}})^{-1}$, whereas the average moisture contents within the cabinet during pull-down, intermittent refrigeration and door opening-closing processes were about $0.263 \text{ g} \cdot (\text{kg}_{\text{dry air}})^{-1}$, $0.315 \text{ g} \cdot (\text{kg}_{\text{dry air}})^{-1}$, and $0.667 \text{ g} \cdot (\text{kg}_{\text{dry air}})^{-1}$. Applying the abovementioned method, the thermal load per second from moist air transfer under both door closing and opening conditions can be determined. Assuming continuous operation of the refrigerator for 24 h, this included a 2.35-h pull-down process, a 21.65-h intermittent refrigeration process and six door opening-closings each lasting 5 s. The accumulative thermal load, equaling to the cold consumption, generated by moist air transfer during these processes has been computed and is presented in Fig. 14.

As depicted in Fig. 14, the thermal load generated during the pull-down process under the door closing condition was slightly higher than that during the intermittent refrigeration process, with value of 3.506 W and 2.693 W. This discrepancy was attributed to the negative pressure during the pull-down process, which led to the continuous intake of moist air, whereas the intermittent refrigeration process involved periodic inhalation and exhalation of air that mitigated the transfer of moist air. The thermal load under the door opening conditions significantly increased to 9412.001 W, with sensible and latent thermal loads accounting for 5219.831 W and 4192.170 W. This indicated that the heat and mass transfer between the inside and outside air of the cabinet when the door was open was intense, producing a thermal load approximately 1500 times higher than under door closing conditions. The sensible thermal load exceeded the latent thermal load, but both were of a similar magnitude, demonstrating that the latent thermal load due to phase change, despite the low water vapor content in the moist air, was significant and cannot be overlooked. Over a 24 h operation cycle, the accumulative thermal loads generated from the pull-down and intermittent refrigeration processes under door closing condition were 29.657 kJ and 209.840 kJ. Although the pull-down process had a higher thermal load per unit time, the intermittent refrigeration process, due to its longer duration, resulted in a greater total thermal load and refrigeration energy consumption. The 30 s door opening-closing processes generated sensible and latent thermal loads as high as 156.595 kJ and 125.765 kJ, surpassing the thermal load under door closing conditions. This indicated that despite the door opening and closing duration accounting for only 0.35 % of the total operation time, it caused the highest thermal load and cold consumption. Within a 24 h period, the cold consumption due to moist air transfer was 417.054 kJ, with the pull-down process, intermittent refrigeration process, and door opening-closing process accounting for 5.7 %, 40.2 %, and 54.1 %.

In addition, to clarify the contribution of thermal load introduced by moist air transfer to the total thermal load of the refrigerator, the total thermal load of Refrigerator A over a 24-h intermittent refrigeration cycle was calculated. This total includes the thermal loads associated with the cabinet walls, the gasket, and the transfer of moist air. Utilizing Eqs. (16) and (17), and the dimensions and materials of the walls and the gasket listed in Table 5, the thermal load of the cabinet walls and gasket were calculated to be 6567.2 kJ and 845.1 kJ. Utilizing Eq. (11)–(14), the thermal load introduced by moist air transfer through the gasket and defrosting drainage tube under door closing conditions were calculated to be 232.0 kJ. With the door opened for 5 s at a 90° angle and a total opening time of 3 min, the thermal loads introduced by moist air transfer were calculated to be 2152.9 kJ. The calculation results reveal that the total thermal load of Refrigerator A over 24 h in intermittent cooling mode is 9797.2 kJ, with the thermal load introduced by moist air transfer amounting to 2384.9 kJ, accounting for approximately 24.3 % of the total thermal load.

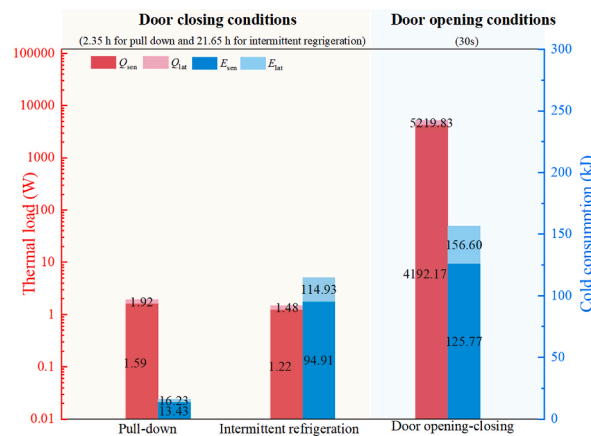


Fig. 14. Comparison of moist air transfer thermal load under door opening and closing conditions.

Table 5

The dimensions and materials of the walls and the gasket.

Positions	Steel		Foam		ABS		Area (m ²)
	δ (mm)	λ (W · m ⁻¹ · K ⁻¹)	δ (mm)	λ (W · m ⁻¹ · K ⁻¹)	δ (mm)	λ (W · m ⁻¹ · K ⁻¹)	
Left and right walls	0.5	38.69	60	0.019	1	0.25	0.997
Front and back walls							1.064
Top and bottom walls							0.378
Gasket	Material: PVC, length: 4.104 m						

7. Conclusion

This study analyzed the transfer characteristics of moist air in refrigerators under door closing and opening conditions and their impact on thermal load. The main findings are as follows:

Under door closing conditions, moist air is transferred through the gasket and defrosting drainage tube. The transfer rate through the defrosting drainage tube is significantly influenced by the design of its structure, accounting for 29.5 % and 88.5 % in the two refrigerators studied in this paper. Furthermore, during the pull-down process and intermittent refrigeration process, the transfer rate of moist air is 133.018 g h⁻¹ and 102.920 g h⁻¹, due to the variation in air pressure difference between the inside and outside of the cabinet.

Under door opening conditions, the transfer rate of moist air varies with the duration and angle of the door opening, ranging from 65.598 to 150.076 g s⁻¹, which is approximately 2300–5200 times higher than the transfer rate under door closing conditions. After the door is closed, the negative pressure formed inside the refrigerator (−100.0 ~ −2000.0 Pa) significantly increases the difficulty of opening the door, with the maximum opening force and secondary opening time reaching up to 1581.0 N and 67 s.

Over a 24 h operating cycle, the cumulative thermal load introduced by moist air transfer is 417.054 kJ, with pull-down, intermittent refrigeration, and door opening-closing processes contributing 5.7 %, 40.2 %, and 54.1 %, indicating that brief door opening processes have a significant impact on the introduction of thermal load. Additionally, the thermal load introduced by moist air transmission accounts for approximately 24.3 % of the total operational thermal load of the refrigerator, with a longer door opening time increasing this proportion.

CRedit authorship contribution statement

Guixiang He: Writing – original draft, Methodology. **Huawei Zhang:** Resources, Conceptualization. **Guoqiang Liu:** Writing – review & editing, Funding acquisition. **Tianyang Zhao:** Methodology, Data curation. **Gang Yan:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work presented in this paper is financially supported by National Natural Science Foundation of China (NSFC) under the grant No. 52306021. The authors would like to thank NSFC for the sponsorship.

Data availability

Data will be made available on request.

References

- [1] UNEP, Refrigeration, Air conditioning and heat pumps technical options committee, 2018 assessment report. https://ozone.unep.org/sites/default/files/2019-04/RTOC-assessment-report-2018_0.pdf, 2019.
- [2] J. Liu, Y. Liu, Experimental investigation of a bypass two-circuit cycle refrigerator-freezer with low GWP zeotropic refrigerants, *Case Stud. Therm. Eng.* 54 (2024) 104071, <https://doi.org/10.1016/j.csite.2024.104071>.
- [3] S. Kaser, R. Nayak, S. Chandra Bhaduri, Performance analysis of solar milk refrigerator using energy efficient R290, *Case Stud. Therm. Eng.* 24 (2021) 100855, <https://doi.org/10.1016/j.csite.2021.100855>.
- [4] N. Nethaji, S.T. Mohideen, Energy conservation in domestic refrigerators by cooling compressor shell – a case study, *Case Stud. Therm. Eng.* 10 (2017) 382–387, <https://doi.org/10.1016/j.csite.2017.08.002>.
- [5] S.A. Borikar, M.M. Gupta, M.A. Alazwari, P.D. Malwe, E.B. Moustafa, H. Panchal, A. Elsheikh, A case study on experimental and statistical analysis of energy consumption of domestic refrigerator, *Case Stud. Therm. Eng.* 28 (2021) 101636, <https://doi.org/10.1016/j.csite.2021.101636>.
- [6] P. Faghihi, A. Salehi, A. Jalali, B. Sajadi, M. Ashjaee, E. Houshfar, Refrigeration components sizing tool for design of domestic refrigerators (ReSiCo): demonstration in full scale, *Case Stud. Therm. Eng.* 49 (2023) 103301, <https://doi.org/10.1016/j.csite.2023.103301>.
- [7] P. Yang, X. Yang, Q. Liu, D. Li, J. Li, G. Yu, Y. Liu, Performance improvement of a household freezer with a microchannel flat-tube evaporator, *Case Stud. Therm. Eng.* 49 (2023) 103394, <https://doi.org/10.1016/j.csite.2023.103394>.
- [8] Y.A. Alghanima, O. Mesalhy, A.F. AbdelGawad, Effect of cold-side heat sink configurations on thermal performance of thermo-electric compartment of a hybrid household refrigerator, *Case Stud. Therm. Eng.* 37 (2022) 102302, <https://doi.org/10.1016/j.csite.2022.102302>.
- [9] G. El Achkar, B. Liu, R. Bennacer, Experimental study on refrigeration performance optimisation of reciprocating room temperature magnetic refrigerator, *Case Stud. Therm. Eng.* 36 (2022) 102224, <https://doi.org/10.1016/j.csite.2022.102224>.
- [10] D.-Y. Liu, W.-R. Chang, J.-Y. Lin, Performance comparison with effect of door opening on variable and fixed frequency refrigerators/freezers, *Appl. Therm. Eng.* 24 (2004) 2281–2292, <https://doi.org/10.1016/j.applthermaleng.2004.01.009>.
- [11] B. Rajkumar, B. Rajkumar, K.R. Aharwal, K.R. Aharwal, Experimental and theoretical analysis of mass transfer in a refrigerated food storage, *Heat Mass Tran.* 58 (2022) 1845–1855, <https://doi.org/10.1007/s00231-022-03217-y>.
- [12] Z. Zhu, W. Dou, B. Dai, X. Shan, J. Shen, Method to delay frost formation under high relative humidity by choosing proper heat exchange temperature difference and air velocity, *Case Stud. Therm. Eng.* 45 (2023) 102922, <https://doi.org/10.1016/j.csite.2023.102922>.
- [13] R.O. Piuco, C.J.L. Hermes, C. Melo, In-situ evaluation of a criterion to predict frost formation on liners of refrigerated cabinets, *Appl. Therm. Eng.* 31 (2011) 3084–3091, <https://doi.org/10.1016/j.applthermaleng.2011.05.003>.
- [14] M. Caciolo, P. Stabat, D. Marchio, Full scale experimental study of single-sided ventilation: analysis of stack and wind effects, *Energy Build.* 43 (2011) 1765–1773, <https://doi.org/10.1016/j.enbuild.2011.03.019>.
- [15] L.J. Lo, A. Novoselac, Cross ventilation with small openings: measurements in a multi-zone test building, *Build. Environ.* 57 (2012) 377–386, <https://doi.org/10.1016/j.applthermaleng.2011.05.039>.
- [16] J. O'Sullivan, M. Ferrua, R. Love, P. Verboven, B. Nicolai, A. East, Airflow measurement techniques for the improvement of forced-air cooling, refrigeration and drying operations, *J. Food Eng.* 143 (2014) 90–101, <https://doi.org/10.1016/j.jfoodeng.2014.06.041>.
- [17] T. Lin, O.A. Zargar, D.L. Sabusap, O. Juina, T.-C. Lee, S.-C. Hu, G. Leggett, Flow characteristics and velocity fields of a front opening unified pod at open door condition, *Case Stud. Therm. Eng.* 21 (2020) 100704, <https://doi.org/10.1016/j.csite.2020.100704>.
- [18] N. Nikolopoulos, A. Nikolopoulos, T.S. Larsen, K.-S.P. Nikas, Experimental and numerical investigation of the tracer gas methodology in the case of a naturally cross-ventilated building, *Build. Environ.* 56 (2012) 379–388, <https://doi.org/10.1016/j.buildenv.2012.04.006>.
- [19] M. Amin, D. Dabiri, H.K. Navaz, Tracer gas technique: a new approach for steady state infiltration rate measurement of open refrigerated display cases, *J. Food Eng.* 92 (2009) 172–181, <https://doi.org/10.1016/j.jfoodeng.2008.10.039>.
- [20] M. Amin, D. Dabiri, H.K. Navaz, Comprehensive study on the effects of fluid dynamics of air curtain and geometry, on infiltration rate of open refrigerated cavities, *Appl. Therm. Eng.* 31 (2011) 3055–3065, <https://doi.org/10.1016/j.applthermaleng.2011.05.039>.
- [21] P.d.V. Pereira, C. Melo, A methodology to measure the rates of air infiltration into refrigerated compartments, *Int. J. Refrig.* 97 (2019) 88–96, <https://doi.org/10.1016/j.jrefrig.2018.08.024>.
- [22] G. Liu, Y. Li, T. Xiong, G. Yan, Study of breathing behavior and its heat load for a refrigerator through the gasket, *Int. Commun. Heat Mass Tran.* 146 (2023) 106944, <https://doi.org/10.1016/j.icheatmasstransfer.2023.106944>.
- [23] G. Liu, G. Yan, J. Yu, A review of refrigerator gasket: development trend, heat and mass transfer characteristics, structure and material optimization, *Renew. Sustain. Energy Rev.* 144 (2021) 110975, <https://doi.org/10.1016/j.rser.2021.110975>.
- [24] C. Afonso, M. Castro, Air infiltration in domestic refrigerators: the influence of the magnetic seals conservation, *Int. J. Refrig.* 33 (2010) 856–867, <https://doi.org/10.1016/j.jrefrig.2009.12.007>.
- [25] S. Kang, D. Kim, S. Lee, D. Kang, E. Yeom, Design of drain tube with movable shutter in household refrigerators, *Int. J. Refrig.* 130 (2021) 76–86, <https://doi.org/10.1016/j.jrefrig.2021.06.001>.
- [26] R. Saidur, H.H. Masjuki, I.A. Choudhury, Role of ambient temperature, door opening, thermostat setting position and their combined effect on refrigerator-freezer energy consumption, *Energy Convers. Manag.* 43 (2002) 845–854, [https://doi.org/10.1016/S0196-8904\(01\)00069-3](https://doi.org/10.1016/S0196-8904(01)00069-3).
- [27] R. Saidur, M.A. Sattar, M. Hasanuzzaman, C.F. Cheng, H.H. Masjuki, Open and closed door moisture transport and corresponding energy consumption in household refrigerator, *Journal of Energy & Environment* 81 6 (2007) 18–27. <https://www.buet.ac.bd/ces/md-hasanuzzaman.pdf>.
- [28] M.S. Aliissi, S. Ramadhyani, R.J. Schoenhals, Effects of ambient temperature, ambient humidity, and door openings on energy consumption of a household refrigerator-freezer, *Build. Eng.* 94 (1988) 1713–1736. <https://iifir.org/en/fridoc/effects-of-ambient-temperature-ambient-humidity-and-door-openings-on-9561>.
- [29] M.F. Vitor, A. dos Santos Silveira, R.C.C. Flesch, Ambient virtual sensor based defrost control for single compartment refrigerators, *Appl. Therm. Eng.* 166 (2020) 114652, <https://doi.org/10.1016/j.applthermaleng.2019.114652>.
- [30] M.A. Stein, C.W. Bullard, T.A. Newell, Moisture transport, frost visualization, and dual evaporator modeling in domestic refrigerators. Air Conditioning and Refrigeration Center, College of Engineering, University of Illinois at Urbana-Champaign, 2000. <http://hdl.handle.net/2142/13352>.
- [31] M.R. Laleman, Sensible and latent energy loading on a refrigerator during open door conditions. Air Conditioning and Refrigeration Center, College of Engineering, University of Illinois at Urbana-Champaign, 1992. <http://hdl.handle.net/2142/9721>.

- [32] D.A. Rodríguez-Valderrama, J.M. Belman-Flores, D. Hernández-Fusilier, D.M. Pardo-Cely, M.A. Gómez-Martínez, S. Méndez-Díaz, Implementation of fuzzy control in a domestic refrigerator considering the influence of the thermal load, *Int. J. Refrig.* 149 (2023) 23–34, <https://doi.org/10.1016/j.ijrefrig.2022.12.018>.
- [33] M. Ipek, I. Dincer, Experimental evaluation of performance parameters of a domestic freezer under door opening conditions, *Therm. Sci. Eng. Prog.* 52 (2024) 102665, <https://doi.org/10.1016/j.tsep.2024.102665>.
- [34] J.A. de Frias, Y. Luo, B. Zhou, B. Zhang, D.T. Ingram, K. Vorst, J.K. Brecht, J. Stommel, Effect of door opening frequency and duration of an enclosed refrigerated display case on product temperatures and energy consumption, *Food Control* 111 (2020), <https://doi.org/10.1016/j.foodcont.2019.107044>.
- [35] H. Tan, X. Song, Y. Zhang, M. He, The mass and heat transfer process through the door seal of refrigeration, *Chin. J. Chem. Eng.* 25 (2017) 1115–1119, <https://doi.org/10.1016/j.cjche.2017.03.016>.
- [36] R.J. Moffat, Describing the uncertainties in experimental results, *Exp. Therm. Fluid Sci.* 1 (1988) 3–17, [https://doi.org/10.1016/0894-1777\(88\)90043-X](https://doi.org/10.1016/0894-1777(88)90043-X).
- [37] T. Zhao, G. Liu, G. He, G. Yan, P. Liu, C. Wang, Effect of structural improvement of gaskets on the heat leakage load and energy consumption of the refrigerator, *Energy* 300 (2024) 131430, <https://doi.org/10.1016/j.energy.2024.131430>.